Prime genealogy in Chebyshev *abc*-orbits: interleaved Lucas atoms and exact radical telescopes

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Working-draft note (2026-07-28): commit `6bf6864` preserves the proposal supplied by Magnus. This version incorporates a Codex proof and software audit plus an independent cross-model review. It remains a non-peer-reviewed research draft.

Abstract

Let ℓ be an odd prime. Splitting the real and imaginary parts of $(\sqrt{b}+i\sqrt{a})^\ell$ gives homogeneous integer polynomials S_ℓ, C_ℓ satisfying

$$aS_\ell(a,b)^2 + bC_\ell(a,b)^2 = (a+b)^\ell.$$

We iterate the resulting degree-ℓ transfer on primitive positive triples $a+b=c$. Under a natural ℓ-adic normalization, every non-seed prime has a unique birth time and a unique branch: the factors

$$A_n = \frac{|S_\ell(a_n,b_n)|}{\ell}, \qquad B_n = |C_\ell(a_n,b_n)|$$

form one pairwise-coprime collection. They are two interleaved towers of homogeneous Lucas atoms, of indices ℓ^{n+1} and $2\ell^{n+1}$, respectively. Writing

$$E_n = A_nB_n, \qquad W_n = \prod_{j<n} \frac{E_j}{\operatorname{rad}(E_j)},$$

we obtain

$$\frac{\operatorname{rad}(a_nb_nc_n)}{c_n} = \frac{\operatorname{rad}(a_0b_0c_0)}{c_0}\,\frac{|\sin(\ell^n\theta)|}{\ell^n \sin\theta\, W_n}, \qquad \cos\theta = \frac{b_0-a_0}{c_0}.$$

Thus every admissible orbit eventually consists of *abc*-hits, the radical ratio decreases strictly at every step, and a persistent quality gap is equivalent to positive-power accumulation in W_n. The factors ℓE_n are successive prime-power cyclotomic factors of a quadratic Lucas sequence. Every prime $p \mid E_n$ has exact order ℓ^{n+1} in the associated norm-one residue group and consequently satisfies

$$\ell^{n+1} \mid p - \left(\frac{D_K}{p}\right).$$

Standard Lucas-atom valuation theory then identifies square divisors of E_n with cyclotomic Wieferich lifts; the orbit identity couples their aggregate multiplicity directly to *abc*-quality. Conversely, the order congruence is locally sufficient: any finite compatible pattern of prime birth levels, branches, split/inert signs, and exact positive multiplicities can be realized by

infinitely many primitive seeds. In particular, the seeds $(\ell, 2, \ell+2)$ give an explicit orbit for every odd prime ℓ, and every member from index 2 onward is an *abc*-hit.

Status. This is a research-development draft, not a peer-reviewed manuscript. The polynomial transfer itself is a classical consequence of de Moivre's formula. The proposed contribution is the orbit-wide package: branchwise prime genealogy in two nested Lucas-atom towers, exact radical identity, monotonicity, the coupling of cyclotomic valuations to *abc*-quality, and simultaneous finite local realization of compatible prime genealogies. A bounded literature search found no source containing this package of statements; that search is not exhaustive.

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Introduction

For a positive integer m, write

$$\operatorname{rad}(m) = \prod_{p \mid m} p.$$

A primitive positive triple (a, b, c) with $a + b = c$ is an *abc*-hit if $\operatorname{rad}(abc) < c$, and its quality is

$$q(a, b, c) = \frac{\log c}{\log \operatorname{rad}(abc)}.$$

Polynomial transfers are a standard way to turn one additive triple into another. Martin and Miao survey several such identities and a general binomial-splitting construction; van der Horst gives a detailed treatment of the transfer method and sharp polynomial triples. Homogeneous cyclotomic factors of Lucas sequences ("Lucas atoms") and their p-adic valuations are also established theory. Ross, Shen, and Cai develop closely related cyclotomic congruences for Lucas sequences. Kym recently obtained valuation separation for products of Lucas terms at pairwise coprime indices.

The purpose here is different: instead of optimizing a single transfer, we identify a prime-degree dynamical orbit whose two coordinate branches are nested-index Lucas-atom towers but whose normalized integer values nevertheless have exactly disjoint support.

Let ℓ be an odd prime and put $m = (\ell - 1)/2$. Define

$$S_\ell(a, b) = \sum_{r=0}^{m} (-1)^r \binom{\ell}{2r+1} a^r b^{m-r},$$
$$C_\ell(a, b) = \sum_{r=0}^{m} (-1)^r \binom{\ell}{2r} a^r b^{m-r}.$$

Then

$$(\sqrt{b} + i\sqrt{a})^\ell = \sqrt{b}\, C_\ell(a, b) + i\sqrt{a}\, S_\ell(a, b),$$

and taking norms gives

$$aS_\ell(a,b)^2 + bC_\ell(a,b)^2 = (a+b)^\ell.$$

The first cases are

$$\begin{aligned} S_3 &= 3b - a, & C_3 &= b - 3a, \\ S_5 &= 5b^2 - 10ab + a^2, & C_5 &= b^2 - 10ab + 5a^2. \end{aligned}$$

Thus the transfer identity includes, for example,

$$a(3b-a)^2 + b(b-3a)^2 = (a+b)^3.$$

The identity is elementary. The arithmetic rigidity appears only after iteration and the normalization at the transfer prime ℓ.

The prime-degree transfer

Let

$$a_0 + b_0 = c_0, \qquad \gcd(a_0, b_0) = 1,$$

where $a_0, b_0 > 0$ have opposite parity. Assume

$$\ell \mid a_0, \qquad \ell \nmid b_0, \qquad v_\ell(S_\ell(a_0, b_0)) = 1.$$

The final valuation condition is automatic when $\ell \geq 5$. For $\ell = 3$, it is the explicit condition

$$v_3(3b_0 - a_0) = 1.$$

Define recursively

$$\begin{aligned} a_{n+1} &= a_n S_n^2, \\ b_{n+1} &= b_n C_n^2, \qquad S_n = S_\ell(a_n, b_n), \quad C_n = C_\ell(a_n, b_n). \\ c_{n+1} &= c_n^\ell, \end{aligned}$$

The transfer identity gives $a_n + b_n = c_n$ for every n, and

$$c_n = c_0^{\ell^n}.$$

Lemma 1: Parity and ℓ-adic normalization

For every $n \geq 0$, the integers a_n, b_n have opposite parity, and

$$\ell \mid a_n, \qquad \ell \nmid b_n c_n, \qquad v_\ell(S_n) = 1, \qquad \ell \nmid C_n.$$

In particular, the integers

$$A_n := \frac{|S_n|}{\ell}, \qquad B_n := |C_n|, \qquad E_n := A_n B_n = \frac{|S_n C_n|}{\ell}$$

are positive and odd, and are coprime to ℓ.

Proof. If a, b have opposite parity, inspection of the definitions of S_ℓ and C_ℓ shows that both $S_\ell(a, b)$ and $C_\ell(a, b)$ are odd. Hence opposite parity is preserved by the orbit.

Suppose $\ell \mid a$ and $\ell \nmid b$. Modulo ℓ,

$$C_\ell(a, b) \equiv b^m \not\equiv 0.$$

For $\ell \geq 5$, divide $S_\ell(a, b)$ by ℓ. All interior binomial coefficients are divisible by ℓ, and the terminal term $(-1)^m a^m/\ell$ is divisible by ℓ because $m \geq 2$. Thus

$$\frac{S_\ell(a, b)}{\ell} \equiv b^m \pmod{\ell},$$

so $v_\ell(S_\ell(a, b)) = 1$.

When $\ell = 3$, the asserted valuation is assumed at $n = 0$. It makes $v_3(a_1) \geq 3$, after which

$$\frac{3b - a}{3} \equiv b \pmod{3},$$

so the valuation remains one. The recurrence now proves all assertions by induction. $\square$

Lemma 2: Primitivity and local support exclusion

Every triple (a_n, b_n, c_n) is primitive. Moreover,

$$\gcd(E_n, a_n b_n c_n) = 1 \qquad (n \geq 0).$$

Proof. Assume inductively that the current triple is primitive; this holds for the seed. Write a, b, c, S, C, E for the corresponding objects, and let q be a prime. In particular, a prime dividing one of a, b, c divides neither of the other two.

If $q \mid a$, then

$$C \equiv b^m \pmod{q}, \qquad S \equiv \ell b^m \pmod{q}.$$

These are nonzero unless $q = \ell$; for $q = \ell$, Lemma 1 shows that the single factor ℓ in S has been removed from E.

If $q \mid b$, then

$$S \equiv (-1)^m a^m \pmod{q}, \qquad C \equiv (-1)^m \ell a^m \pmod{q},$$

and $q \neq \ell$ by Lemma 1.

If $q \mid c = a + b$, then $b \equiv -a \pmod{q}$. Since c is odd, q is odd, and direct summation of the even and odd binomial coefficients gives

$$S_\ell(a, -a) = C_\ell(a, -a) = (-1)^m 2^{\ell-1} a^m.$$

This is nonzero modulo q. These congruences prove the local support exclusion.

It remains to rule out a prime common to the two new summands aS^2 and bC^2. A common prime cannot arise from an old factor by the congruences above. If it divided both S and C, then the transfer identity would force it to divide c, contradicting the boundary evaluation; the prime 2 is excluded because S, C are odd.

Hence the transferred triple is primitive, and induction completes the proof. □

Theorem 3: Branchwise prime genealogy

The full collection

$$A_0, B_0, A_1, B_1, A_2, B_2, \ldots$$

is pairwise coprime, and every member is coprime to $a_0 b_0 c_0$. More precisely,

$$\begin{aligned} a_n &= a_0 \ell^{2n} \prod_{j=0}^{n-1} A_j^2, \\ b_n &= b_0 \prod_{j=0}^{n-1} B_j^2, \\ c_n &= c_0^{\ell^n}. \end{aligned}$$

If

$$R_n = \operatorname{rad}(a_n b_n c_n),$$

then

$$\boxed{R_n = R_0 \prod_{j=0}^{n-1} \operatorname{rad}(A_j B_j).}$$

Thus every non-seed prime has a unique birth generation and a unique coordinate branch. Its valuation in that coordinate is fixed forever after birth.

Proof. Every prime of A_j divides a_{j+1}, and every prime of B_j divides b_{j+1}; hence either prime divides $a_n b_n$ for every $n > j$. Lemma 2 then excludes it from both A_n and B_n.

At a fixed generation, a prime common to A_n and B_n would divide both S_n and C_n, which was excluded in the proof of Lemma 2. The same local equation at $n = 0$ gives coprimality with the seed. This proves the pairwise claim.

Since

$$S_j = \pm \ell A_j, \qquad C_j = \pm B_j,$$

iterating the orbit gives the displayed coordinate factorizations. The factors are support-disjoint, so taking radicals gives the radical factorization. Because later multipliers are coprime to every earlier factor, the valuation of a born prime never changes. □

Chebyshev dynamics and the radical telescope

Set

$$x_n = \frac{b_n - a_n}{c_n}, \qquad x_0 = \cos\theta, \qquad 0 < \theta < \pi.$$

The angle exists because $-1 < x_0 < 1$, and

$$\sin\theta = \frac{2\sqrt{a_0 b_0}}{c_0}.$$

Lemma 4: Chebyshev semiconjugacy

Let T_r, U_r be the Chebyshev polynomials of the first and second kind. Then

$$\begin{aligned} x_{n+1} &= T_\ell(x_n), \\ x_n &= T_{\ell^n}(x_0) = \cos(\ell^n\theta), \end{aligned}$$

and

$$\frac{S_n C_n}{c_n^{\ell-1}} = U_{\ell-1}(x_n) = \frac{\sin(\ell^{n+1}\theta)}{\sin(\ell^n\theta)}.$$

None of the sine values in the final expression vanishes.

Proof. Choose ϕ_n with

$$\sin^2\phi_n = \frac{a_n}{c_n}, \qquad \cos^2\phi_n = \frac{b_n}{c_n},$$

so that

$$x_n = \cos(2\phi_n).$$

The real and imaginary parts in the definition of S_ℓ, C_ℓ give

$$\frac{S_n}{c_n^m} = \frac{\sin(\ell\phi_n)}{\sin\phi_n}, \qquad \frac{C_n}{c_n^m} = \frac{\cos(\ell\phi_n)}{\cos\phi_n}.$$

Therefore the normalized new coordinates are

$$\frac{a_{n+1}}{c_{n+1}} = \sin^2(\ell\phi_n), \qquad \frac{b_{n+1}}{c_{n+1}} = \cos^2(\ell\phi_n),$$

proving the Chebyshev dynamics. Multiplying the displayed formulas and using the double-angle identity gives the $U_{\ell-1}$ formula.

If a sine vanished, then $z = e^{i\theta}$ would be a root of unity. But

$$z + z^{-1} = 2\frac{b_0 - a_0}{c_0}$$

would then be a rational algebraic integer. Since

$$\gcd(c_0, b_0 - a_0) = 1,$$

while $c_0 > 1$ is odd, this is impossible. $\square$

Define

$$W_n = \prod_{j=0}^{n-1} \frac{E_j}{\operatorname{rad}(E_j)}.$$

Theorem 5: Exact prime-degree radical identity

For every $n \geq 0$,

$$\boxed{\frac{R_n}{c_n} = \frac{R_0}{c_0}\,\frac{|\sin(\ell^n\theta)|}{\ell^n \sin\theta\, W_n}}.$$

Consequently,

$$\frac{c_n}{R_n} \geq \frac{2\sqrt{a_0 b_0}}{R_0}\,\ell^n = \frac{2\sqrt{a_0 b_0}}{R_0 \log c_0}\log c_n.$$

Every admissible orbit eventually consists of *abc*-hits.

Proof. Since

$$\prod_{j<n} c_j^{\ell-1} = c_0^{\ell^n-1} = \frac{c_n}{c_0},$$

Theorem 3 and the definitions of E_n, W_n give

$$\frac{R_n}{c_n} = \frac{R_0}{c_0}\frac{1}{\ell^n W_n}\prod_{j<n}\frac{|S_j C_j|}{c_j^{\ell-1}}.$$

Lemma 4 telescopes the product to

$$\frac{|\sin(\ell^n\theta)|}{\sin\theta},$$

proving the identity. Now use

$$W_n \geq 1, \qquad |\sin(\ell^n\theta)| \leq 1, \qquad \sin\theta = \frac{2\sqrt{a_0 b_0}}{c_0}$$

to obtain the lower bound. □

Proposition 6: Strict radical monotonicity

The normalized radical decreases at every transfer:

$$\frac{R_{n+1}}{c_{n+1}} < \frac{R_n}{c_n} \qquad (n \geq 0).$$

In particular, once an orbit member is an *abc*-hit, every later member is a hit.

Proof. Theorem 3 and the growth of c_n give

$$\frac{R_{n+1}/c_{n+1}}{R_n/c_n} = \frac{\operatorname{rad}(E_n)}{c_n^{\ell-1}} \leq \frac{|U_{\ell-1}(x_n)|}{\ell}.$$

For $-1 < x < 1$, writing $x = \cos t$ gives

$$|U_{\ell-1}(x)| = \frac{|\sin(\ell t)|}{\sin t} < \ell.$$

The inequality is strict away from $t = 0, \pi$, which do not occur here. □

Effective quality control

The exact identity isolates the only archimedean oscillation. It is much smaller than $\log c_n$.

Theorem 7: Archimedean term

There is an effective constant

$$A = A(a_0, b_0, \ell) > 0$$

such that

$$|\sin(\ell^n \theta)| > \exp(-An) \qquad (n \geq 1).$$

Consequently,

$$\log \frac{c_n}{R_n} = \log W_n + \Theta_{a_0,b_0,\ell}(n).$$

Proof. Let

$$z = e^{i\theta} = \frac{b_0 - a_0 + 2\sqrt{-a_0 b_0}}{c_0}.$$

Lemma 4 shows that z is not a root of unity. Put

$$N = \ell^n,$$

choose an integer k nearest to $N\theta/\pi$, and use principal logarithms to form

$$\Lambda_N = N \log z - k \log(-1) = i(N\theta - k\pi) \neq 0.$$

The explicit theorem on linear forms in logarithms of Baker and Wüstholz, applied to z and -1, gives

$$\log |\Lambda_N| > -C \log N$$

for an effective seed-dependent constant C.

Since

$$|N\theta - k\pi| \leq \frac{\pi}{2},$$

the elementary inequality

$$|\sin y| \geq \frac{2|y|}{\pi}$$

on that interval gives

$$|\sin(N\theta)| \gg N^{-C} = \exp(-Cn\log\ell).$$

Taking logarithms in the exact radical identity now gives

$$\log\frac{c_n}{R_n} = \log W_n + n\log\ell + \log\frac{c_0\sin\theta}{R_0} - \log|\sin(\ell^n\theta)|.$$

The lower estimate follows from $|\sin| \le 1$, and the upper estimate follows from the Baker-Wüstholz bound. □

Corollary 8: Exact quality criterion

Put

$$q_n = \frac{\log c_n}{\log R_n}.$$

Then

$$q_n \longrightarrow 1 \quad \Longleftrightarrow \quad \log W_n = o(\log c_n),$$

and

$$\limsup_{n\to\infty}(q_n - 1) > 0 \quad \Longleftrightarrow \quad \limsup_{n\to\infty}\frac{\log W_n}{\log c_n} > 0.$$

For every fixed $\delta > 0$, if

$$F_n = \log(c_n/R_n) - \log W_n,$$

then

$$q_n \ge 1+\delta \quad \Longleftrightarrow \quad \log W_n + F_n \ge \frac{\delta}{1+\delta}\log c_n.$$

Proof. By Theorem 5, one has $R_n < c_n$ for all sufficiently large n. Write

$$\Delta_n = \log(c_n/R_n).$$

Since

$$\log c_n = \ell^n \log c_0,$$

Theorem 7 gives

$$\Delta_n = \log W_n + o(\log c_n).$$

Now

$$q_n - 1 = \frac{\Delta_n}{\log R_n}.$$

This proves the two asymptotic equivalences, and direct rearrangement gives the exact threshold. $\square$

Cyclotomic Lucas factors and level primes

The support separation has a more refined interpretation than the product $E_n = A_n B_n$. Put

$$\omega = \sqrt{b_0} + \sqrt{-a_0}, \qquad \bar{\omega} = \sqrt{b_0} - \sqrt{-a_0}.$$

Proposition 9: Two interleaved Lucas-atom towers

For every $n \geq 0$,

$$S_n = \Phi_{\ell^{n+1}}(\omega, \bar{\omega}), \qquad C_n = \Phi_{2\ell^{n+1}}(\omega, \bar{\omega}).$$

Consequently, up to the systematic factor ℓ in the first tower, the a-branch and b-branch of the orbit are the nested-index Lucas atoms ℓ^{n+1} and $2\ell^{n+1}$, respectively.

Proof. Let

$$z_n = \omega^{\ell^n} = r_n + s_n\sqrt{-1},$$

under the fixed complex embedding. De Moivre's formula and induction in the orbit give

$$r_n^2 = b_n, \qquad s_n^2 = a_n.$$

Therefore

$$S_n = \frac{z_n^\ell - \bar{z}_n^\ell}{z_n - \bar{z}_n}, \qquad C_n = \frac{z_n^\ell + \bar{z}_n^\ell}{z_n + \bar{z}_n}.$$

Substituting $z_n = \omega^{\ell^n}$ and using the prime-power factorizations

$$\frac{X^{\ell^{n+1}} - Y^{\ell^{n+1}}}{X^{\ell^n} - Y^{\ell^n}} = \Phi_{\ell^{n+1}}(X, Y),$$

and

$$\frac{X^{\ell^{n+1}} + Y^{\ell^{n+1}}}{X^{\ell^n} + Y^{\ell^n}} = \Phi_{2\ell^{n+1}}(X,Y)$$

proves the claim. □

Let

$$L = \mathbb{Q}(\sqrt{b_0}, \sqrt{-a_0})$$

and

$$v = \frac{\omega}{\bar{\omega}}.$$

The branch of a prime records a sharper order condition than the product alone.

Proposition 10: Branch order

Let p be a rational prime and $\mathfrak{P} \mid p$ in L. Then

$$p \mid A_n \quad \Longrightarrow \quad \operatorname{ord}(\widetilde{v}) = \ell^{n+1},$$

and

$$p \mid B_n \quad \Longrightarrow \quad \operatorname{ord}(\widetilde{v}) = 2\ell^{n+1}.$$

Here $\widetilde{v}$ is the reduction of v modulo $\mathfrak{P}$. Thus the unique coordinate branch in Theorem 3 is also a unique cyclotomic order label.

Proof. Theorem 3 excludes

$$p \mid 2\ell a_0 b_0 c_0,$$

so ω and $\bar{\omega}$ are units modulo $\mathfrak{P}$. Divide the atom formulas by the appropriate power of $\bar{\omega}$. Since the residue characteristic does not divide either cyclotomic index, a zero of $\Phi_{\ell^{n+1}}$, respectively $\Phi_{2\ell^{n+1}}$, has exactly the displayed order. □

Proposition 11: Signed branch Wieferich lifts

If $p \mid A_n$, respectively $p \mid B_n$, then

$$p^2 \mid A_n \quad \Longleftrightarrow \quad v^{\ell^{n+1}} \equiv 1 \pmod{\mathfrak{P}^2},$$

and

$$p^2 \mid B_n \quad \Longleftrightarrow \quad v^{\ell^{n+1}} \equiv -1 \pmod{\mathfrak{P}^2}.$$

Thus a repeated prime records not only its level but also, through the sign of the lift, the coordinate in which it was born.

Proof. For the first branch, the exact-order statement makes $v^{\ell^n} - 1$ a $\mathfrak{P}$-adic unit, while

$$\Phi_{\ell^{n+1}}(v) = \frac{v^{\ell^{n+1}} - 1}{v^{\ell^n} - 1}.$$

For the second branch, v^{ℓ^n} has order 2ℓ, so $v^{\ell^n} + 1$ is also a unit, and

$$\Phi_{2\ell^{n+1}}(v) = \frac{v^{\ell^{n+1}} + 1}{v^{\ell^n} + 1}.$$

Indeed, Theorem 3 gives $p \nmid 2a_0b_0$. Every odd prime ramified in $\mathbb{Q}(\sqrt{b_0})$, respectively $\mathbb{Q}(\sqrt{-a_0})$, divides b_0, respectively a_0. Both quadratic extensions are therefore unramified at p, and so is their compositum L. Since A_n, B_n are rational integers and the ramification index is one, their p-adic valuations equal their $\mathfrak{P}$-adic valuations. Hence divisibility by p^2 is equivalent to valuation at least two at $\mathfrak{P}$, and the two equivalences follow. □

Squaring ω descends the two branches to one quadratic Lucas atom. Define

$$\alpha = b_0 - a_0 + 2\sqrt{-a_0b_0}, \qquad \beta = b_0 - a_0 - 2\sqrt{-a_0b_0}.$$

Then

$$\alpha\beta = c_0^2.$$

Let

$$\mathcal{U}_r = \frac{\alpha^r - \beta^r}{\alpha - \beta} \qquad (r \geq 1)$$

be the associated quadratic Lucas sequence, and let

$$\Phi_r(X, Y) = Y^{\varphi(r)}\Phi_r(X/Y)$$

denote the homogeneous cyclotomic polynomial.

Proposition 12: Prime-power cyclotomic factorization

For every $n \geq 0$,

$$S_nC_n = \frac{\mathcal{U}_{\ell^{n+1}}}{\mathcal{U}_{\ell^n}} = \Phi_{\ell^{n+1}}(\alpha, \beta).$$

Thus ℓE_n is the absolute value of the ℓ^{n+1}-st homogeneous cyclotomic factor of the fixed Lucas pair (α, β).

Proof. Write

$$\alpha = c_0 e^{i\theta}, \qquad \beta = c_0 e^{-i\theta}.$$

Then

$$\mathcal{U}_r = c_0^{r-1} \frac{\sin(r\theta)}{\sin\theta}.$$

Taking the quotient at $r = \ell^{n+1}$ and $r = \ell^n$, and using $c_n = c_0^{\ell^n}$, gives the formula from Lemma 4. The second equality is the standard identity

$$\frac{X^{\ell^{n+1}} - Y^{\ell^{n+1}}}{X^{\ell^n} - Y^{\ell^n}} = \Phi_{\ell^{n+1}}(X, Y).$$

□

Let

$$K = \mathbb{Q}(\sqrt{-a_0 b_0}),$$

let D_K be its field discriminant, and put

$$u = \frac{\alpha}{\beta} \in K^\times.$$

The norm of u is one.

Theorem 13: Exact level and prime congruence

Let $p \mid E_n$ be a rational prime, and let $\mathfrak{p} \mid p$ in K. Then

$$p \nmid 2\ell a_0 b_0 c_0,$$

and the image $\widetilde{u}$ of u in $(\mathcal{O}_K/\mathfrak{p})^\times$ has exact order

$$\mathrm{ord}(\widetilde{u}) = \ell^{n+1}.$$

Consequently,

$$\boxed{\ell^{n+1} \mid p - \left(\frac{D_K}{p}\right).}$$

In particular, every new prime introduced at generation n lies in one of the two residue classes

$$\pm 1 \pmod{\ell^{n+1}},$$

with the sign determined by splitting in K.

Proof. The exclusion of $2, \ell$, and the seed primes follows from Theorem 3. Hence p is unramified in K, and $\alpha, \beta, \alpha - \beta$ are units modulo $\mathfrak{p}$.

By Proposition 12,

$$\Phi_{\ell^{n+1}}(u) \equiv 0 \pmod{\mathfrak{p}}.$$

Since $p \neq \ell$, a root of this cyclotomic polynomial in the residue field has exact order ℓ^{n+1}.

If p splits in K, then

$$\widetilde{u} \in \mathbb{F}_p^\times,$$

so its order divides $p-1$. If p is inert, Frobenius acts as conjugation and therefore

$$\widetilde{u}^p = \widetilde{\overline{u}} = \widetilde{u}^{-1};$$

its order divides $p+1$. These two cases give the displayed congruence. □

Proposition 14: Cyclotomic Wieferich criterion

Let $p \mid E_n$, and let $\mathfrak{p} \mid p$ in K. Then

$$p^2 \mid E_n \quad \Longleftrightarrow \quad u^{\ell^{n+1}} \equiv 1 \pmod{\mathfrak{p}^2}.$$

The condition is independent of the choice of $\mathfrak{p} \mid p$. Moreover,

$$W_n = \prod_{j=0}^{n-1} \prod_{p \mid E_j} p^{v_p(E_j)-1}.$$

Thus the nonarchimedean term governing the asymptotic *abc*-quality is exactly the aggregate multiplicity of cyclotomic Wieferich lifts at ℓ-power levels. The local valuation theory of Lucas atoms is known in substantially greater generality; the point here is its exact identification with the global defect term W_n of a single additive orbit.

Proof. Modulo $\mathfrak{p}$, the element u^{ℓ^n} is not one by Theorem 13. Hence the denominator in

$$\Phi_{\ell^{n+1}}(u) = \frac{u^{\ell^{n+1}} - 1}{u^{\ell^n} - 1}$$

is a $\mathfrak{p}$-adic unit. Therefore the valuation of the cyclotomic factor is at least two exactly when the numerator vanishes modulo $\mathfrak{p}^2$.

Since E_n is rational, conjugate prime ideals above a split prime occur with equal valuations. The product formula for W_n is its definition expanded prime by prime. □

Local classification and exact realization of prime genealogies

The congruence in Theorem 13 is not only necessary. When the seed is allowed to vary, it is the complete local compatibility condition for these branch polynomials, and the valuation can be prescribed exactly.

Define the coordinate polynomials recursively from $a_0(X,Y) = X$, $b_0(X,Y) = Y$ by the same transfer. Induction shows that a_n, b_n are homogeneous of degree ℓ^n. Hence the level-n multipliers $S_n(X,Y), C_n(X,Y)$ are homogeneous of degree

$$d_n = \frac{\ell^n(\ell-1)}{2}.$$

For a prime $p \neq \ell$, write over $\mathbb{Z}_{(p)}$

$$\mathcal{F}_{n,A}(X,Y) = \frac{S_n(X,Y)}{\ell}, \qquad \mathcal{F}_{n,B}(X,Y) = C_n(X,Y),$$

where $S_n(X,Y), C_n(X,Y)$ mean the multipliers at level n in the orbit started from (X,Y). Put

$$m_{n,A} = \ell^{n+1}, \qquad m_{n,B} = 2\ell^{n+1},$$

For later reduction modulo p, we record why the atom identities are universal polynomial identities rather than identities tied to one complex embedding. Work in the quadratic algebra

$$\mathcal{R} = \mathbb{Z}[X,Y,r,t]/(r^2 - Y,\ t^2 + X)$$

and put $\Omega = r + t$, $\overline{\Omega} = r - t$. Direct binomial expansion gives

$$\Omega^\ell = rC_\ell(X,Y) + tS_\ell(X,Y).$$

Write $\Omega^{\ell^n} = r_n + t_n$, where the involution $(r,t) \mapsto (r,-t)$ fixes r_n and negates t_n. Applying the displayed binomial identity at $(a,b) = (-t_n^2, r_n^2)$ gives

$$r_{n+1} = r_nC_n(X,Y), \qquad t_{n+1} = t_nS_n(X,Y).$$

Since $r_0^2 = Y = b_0(X,Y)$ and $t_0^2 = -X = -a_0(X,Y)$, induction gives $r_n^2 = b_n(X,Y)$ and $t_n^2 = -a_n(X,Y)$. Consequently, in $\mathcal{R}$,

$$S_n(X,Y) = \frac{\Omega^{\ell^{n+1}} - \overline{\Omega}^{\ell^{n+1}}}{\Omega^{\ell^n} - \overline{\Omega}^{\ell^n}} = \Phi_{\ell^{n+1}}(\Omega, \overline{\Omega}),$$
$$C_n(X,Y) = \frac{\Omega^{\ell^{n+1}} + \overline{\Omega}^{\ell^{n+1}}}{\Omega^{\ell^n} + \overline{\Omega}^{\ell^n}} = \Phi_{2\ell^{n+1}}(\Omega, \overline{\Omega}).$$

The quotients denote the corresponding polynomial factorizations, so no inversion is being asserted in $\mathcal{R}$. These identities therefore survive every specialization of $\mathcal{R}$, including the split and quadratic residue algebras used below. In particular, $\deg \mathcal{F}_{n,\varepsilon} = d_n$, and $\varphi(m_{n,\varepsilon}) = 2d_n$ for either branch.

Proposition 15: Local root classification

Fix $n \geq 0$, a branch $\varepsilon \in \{A, B\}$, and an odd prime $p \neq \ell$. Let

$$m = m_{n,\varepsilon}.$$

If

$$m \mid p - \chi \qquad \text{for some } \chi \in \{1, -1\},$$

then $\mathcal{F}_{n,\varepsilon}(X, 1)$ has exactly d_n distinct, simple roots in $\mathbb{F}_p$. They are

$$\rho_\zeta = -\left(\frac{\zeta - 1}{\zeta + 1}\right)^2,$$

where ζ ranges, modulo $\zeta \sim \zeta^{-1}$, over the elements of exact order m in $\mathbb{F}_p^\times$ when $\chi = 1$, and over the norm-one subgroup of $\mathbb{F}_{p^2}^\times$ when $\chi = -1$.

Every such root satisfies

$$\rho_\zeta \neq 0, -1, \qquad \left(\frac{-\rho_\zeta}{p}\right) = \chi.$$

Conversely, every root arises in this way. Hence roots exist if and only if

$$\ell^{n+1} \mid p - 1 \quad \text{or} \quad \ell^{n+1} \mid p + 1.$$

Proof. Under the compatibility condition, choose ζ of exact order m in the indicated cyclic group. Such an element exists because both groups are cyclic. Put

$$s = \frac{\zeta - 1}{\zeta + 1}, \qquad \rho = -s^2.$$

If $\chi = 1$, then $s, \rho \in \mathbb{F}_p$.

If $\chi = -1$, Frobenius on the norm-one group gives

$$\zeta^p = \zeta^{-1},$$

hence

$$s^p = -s$$

and again $\rho \in \mathbb{F}_p$.

In either case, work in $\mathbb{F}_p$ when $-\rho$ is a square and in $\mathbb{F}_{p^2}$ otherwise. The substitutions

$$\sqrt{b_0} \longmapsto 1, \qquad \sqrt{-a_0} \longmapsto s$$

for $(a_0, b_0) = (\rho, 1)$ give

$$\frac{\omega}{\bar{\omega}} \longmapsto \frac{1+s}{1-s} = \zeta.$$

Proposition 9 therefore makes the selected branch atom vanish. The elements s, $1-s$, and $1+s$ are nonzero, so

$$\rho \neq 0, -1.$$

In the split case,

$$-\rho = s^2$$

is a square. In the inert case it is a nonsquare: if s^2 were a square in $\mathbb{F}_p$, then s itself would lie in $\mathbb{F}_p$, forcing

$$\zeta = \frac{1+s}{1-s}$$

into $\mathbb{F}_p$. This is impossible because an odd ℓ-power divides m, whereas $\gcd(p-1, p+1) = 2$; thus the compatible sign χ is unique and $m \nmid p-1$ in the inert case. This proves the Legendre-symbol assertion.

Moreover,

$$\mathcal{F}_{n,\varepsilon}(0,1) = 1,$$

while the atom evaluation at $(X, Y) = (-1, 1)$, corresponding to $(\omega, \bar{\omega}) = (2, 0)$, gives

$$\mathcal{F}_{n,A}(-1,1) = \ell^{-1} 2^{\varphi(m_{n,A})}, \qquad \mathcal{F}_{n,B}(-1,1) = 2^{\varphi(m_{n,B})}$$

in $\mathbb{Z}_{(p)}$. Neither exceptional projective point is therefore a root modulo the odd prime $p \neq \ell$.

The $\varphi(m)$ elements of exact order m yield exactly

$$\frac{\varphi(m)}{2} = d_n$$

values of ρ, because

$$\rho_\zeta = \rho_{\zeta^{-1}},$$

and equality of two values forces the corresponding Cayley parameters to differ by sign. Since $\deg \mathcal{F}_{n,\varepsilon}(X,1) \le d_n$, these d_n roots exhaust the polynomial and are simple.

Conversely, let ρ be any root. The exceptional values just calculated show that $\rho \ne 0, -1$. Choose s with $s^2 = -\rho$ and set $\zeta = (1+s)/(1-s)$. Dividing the homogeneous atom by the appropriate power of $\bar{\omega} = 1 - s$ shows that $\Phi_m(\zeta) = 0$. Since $p \nmid m$, the element ζ has exact order m. If $-\rho$ is a square, then $\zeta \in \mathbb{F}_p^\times$, so $m \mid p-1$. Otherwise $s^p = -s$, hence $\zeta^p = \zeta^{-1}$; thus ζ lies in the norm-one group and $m \mid p+1$. This proves the converse and the Legendre-symbol classification.

For the B-branch the factor 2 in m imposes no extra congruence, because $p - \chi$ is already even. □

Corollary 16: Exact local valuation

Every root ρ in Proposition 15 has a unique lift in its residue class,

$$\widehat{\rho} \in \mathbb{Z}_p$$

satisfying

$$\mathcal{F}_{n,\varepsilon}(\widehat{\rho}, 1) = 0.$$

For every $h \ge 1$ and $\lambda \in \mathbb{Z}_p^\times$,

$$v_p\left(\mathcal{F}_{n,\varepsilon}(\widehat{\rho} + \lambda p^h, 1)\right) = h.$$

Proof. The root is simple, so Hensel's lemma gives the unique lift in $\rho + p\mathbb{Z}_p$, and the derivative at that lift is a p-adic unit. Taylor expansion at $\widehat{\rho}$ gives

$$\mathcal{F}_{n,\varepsilon}(\widehat{\rho} + \lambda p^h, 1) \equiv \lambda p^h \mathcal{F}'_{n,\varepsilon}(\widehat{\rho}, 1) \pmod{p^{2h}}.$$

The displayed main term has valuation h, whereas $2h \ge h+1$. The valuation is therefore exactly h. □

Theorem 17: Simultaneous finite genealogy realization

Fix an odd prime ℓ. Let

$$\mathcal{D} = \{(p_i, n_i, \varepsilon_i, h_i, \chi_i) : 1 \le i \le r\}$$

be finite data in which:

- the p_i are distinct odd primes different from ℓ;
- $n_i \geq 0$;
- $h_i \geq 1$;
- $\varepsilon_i \in \{A, B\}$;
- $\chi_i \in \{1, -1\}$.

Suppose

$$m_{n_i,\varepsilon_i} \mid p_i - \chi_i.$$

Then there are infinitely many admissible primitive seeds for which

$$\varepsilon_i = A \quad \Longrightarrow \quad v_{p_i}(A_{n_i}) = h_i,$$

$$\varepsilon_i = B \quad \Longrightarrow \quad v_{p_i}(B_{n_i}) = h_i,$$

$$\left(\frac{D_K}{p_i}\right) = \chi_i,$$

and

$$p_i \nmid a_0 b_0 c_0,$$

and

$$p_i \nmid A_j B_j \qquad (0 \leq j < n_i),$$

where

$$K = \mathbb{Q}(\sqrt{-a_0 b_0}).$$

Hence every p_i is born at exactly the prescribed generation, in exactly the prescribed branch, with exactly the prescribed multiplicity and splitting sign. It never occurs in a later generation factor.

Proof. For each datum choose a simple root ρ_i from Proposition 15, let $\widehat{\rho}_i$ be its p_i-adic lift, and choose an integer r_i representing $\widehat{\rho}_i$ modulo $p_i^{h_i+1}$. Impose

$$b_0 \equiv 1 \pmod{p_i^{h_i+1}},$$

and

$$a_0 \equiv r_i + p_i^{h_i} \pmod{p_i^{h_i+1}}.$$

Let $d = d_{n_i}$ and $f_i(T) = \mathcal{F}_{n_i,\varepsilon_i}(T, 1)$. Homogeneity gives

$$\mathcal{F}_{n_i,\varepsilon_i}(a_0, b_0) = b_0^d f_i(a_0/b_0).$$

The imposed congruences imply

$$\frac{a_0}{b_0} \equiv \widehat{\rho}_i + p_i^{h_i} \pmod{p_i^{h_i+1}}.$$

The factor b_0^d is a p_i-adic unit, so Corollary 16 gives the exact prescribed valuation.

Reduction modulo p_i leaves the exact order m_{n_i,ε_i} unchanged. Earlier A- and B-atoms have orders ℓ^{j+1} and $2\ell^{j+1}$ with $j < n_i$, and the opposite target atom has the other one of $\ell^{n_i+1}, 2\ell^{n_i+1}$. None equals the prescribed exact order. Thus every earlier atom and the opposite atom at the target level are nonzero. Proposition 15 gives

$$\left(\frac{-a_0 b_0}{p_i}\right) = \chi_i,$$

and

$$a_0 + b_0 \equiv \rho_i + 1 \not\equiv 0 \pmod{p_i}.$$

Thus $p_i \nmid a_0 b_0 c_0$. The field discriminant D_K differs from the squarefree representative of $-a_0 b_0$ only by a rational square and possibly a factor 4. Therefore the preceding symbol equals $\left(\frac{D_K}{p_i}\right)$, because p_i is odd.

This proves all local assertions.

It remains to realize them simultaneously by primitive positive integers. Combine the local congruences by the Chinese remainder theorem and also require a_0 even and b_0 odd.

For $\ell \geq 5$, impose

$$a_0 \equiv \ell \pmod{\ell^2}, \qquad b_0 \equiv 1 \pmod{\ell^2}.$$

For $\ell = 3$, impose

$$a_0 \equiv 3 \pmod 9, \qquad b_0 \equiv 2 \pmod 9.$$

The latter gives

$$v_3(3b_0 - a_0) = 1.$$

All moduli are pairwise coprime, so the Chinese remainder theorem gives classes $\bar{a}, \bar{b} \pmod M$, where

$$M = 2\ell^2 \prod_{i=1}^{r} p_i^{h_i+1}.$$

The class $\bar{b}$ is a unit modulo every factor of M, hence modulo M. Choose infinitely many positive integers $b_0 \equiv \bar{b} \pmod{M}$. Then $\gcd(b_0, M) = 1$, so a second application of the Chinese remainder theorem supplies a positive a_0 satisfying

$$a_0 \equiv \bar{a} \pmod{M}$$

and

$$a_0 \equiv 1 \pmod{b_0},$$

simultaneously. Thus

$$\gcd(a_0, b_0) = 1,$$

and the parity and ℓ-adic conditions give admissibility. Distinct choices of b_0 give distinct seeds. Finally, Theorem 3 excludes every prescribed prime from all later generation factors. $\square$

Remark 18: A programmed square

For $\ell = 3$, the admissible seed

$$(a_0, b_0) = (304{,}260{,}006, 39{,}305)$$

satisfies

$$17 \nmid A_0 B_0, \qquad v_{17}(A_1) = 2, \qquad \left(\frac{D_K}{17}\right) = -1.$$

Thus 17 is programmed to be born in the A-branch at level 1, with multiplicity exactly two and inert sign. The accompanying verification script constructs this seed from the local root and checks the assertions.

Corollary 19: Unbounded fixed-level defect across seeds

Fix ℓ, a generation n, and either branch. For every odd prime $p \neq \ell$ satisfying

$$\ell^{n+1} \mid p - 1 \quad \text{or} \quad \ell^{n+1} \mid p + 1,$$

and every $H \geq 1$, there are infinitely many admissible seeds for which the selected generation factor has p-adic valuation exactly H.

In particular, W_{n+1} is unbounded as the seed varies, even with ℓ and n fixed.

This does not address Conjecture 21, which fixes one seed and lets n tend to infinity. Rather, it shows that no uniform squarefreeness or valuation bound can hold across all seeds. More sharply, the necessary split/inert congruence and the signed branch conditions are locally sufficient at every level, with arbitrary exact multiplicity.

A canonical family for every odd prime

Corollary 20: Uniform explicit hit orbits

For every odd prime ℓ, the seed

$$(a_0, b_0, c_0) = (\ell, 2, \ell + 2)$$

is admissible. Every member of its orbit with $n \geq 2$ is an abc-hit.

Proof. The seed is primitive and has opposite parity. For $\ell \geq 5$, admissibility follows from Lemma 1. For $\ell = 3$, one has

$$S_3(3, 2) = 3.$$

Since

$$R_0 = 2\ell \; \mathrm{rad}(\ell + 2) \leq 2\ell(\ell + 2),$$

Theorem 5 gives

$$\frac{c_n}{R_n} \geq \frac{\ell^{n-1}\sqrt{2\ell}}{\ell + 2}.$$

At $n = 2$, the right side is greater than one for every odd prime $\ell \geq 3$, and it increases with n. $\square$

The first two nontrivial examples illustrate that the guaranteed hit need not occur at the first transfer.

The cubic orbit

For $\ell = 3$,

$$(3, 2, 5) \longmapsto (27, 98, 125) \longmapsto (1{,}924{,}803, 28{,}322, 1{,}953{,}125).$$

The second transferred triple factors as

$$\begin{aligned} 1{,}924{,}803 &= 3^5 \cdot 89^2, \\ 28{,}322 &= 2 \cdot 7^2 \cdot 17^2, \\ 1{,}953{,}125 &= 5^9, \end{aligned}$$

so

$$R_2 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 17 \cdot 89 = 317{,}730,$$

and

$$\frac{c_2}{R_2} = 6.1471\ldots$$

The first two generation factors are

$$E_0 = 7, \qquad E_1 = 17 \cdot 89.$$

Their prime divisors satisfy respectively

$$p \equiv \pm 1 \pmod 3$$

and

$$p \equiv \pm 1 \pmod 9,$$

as predicted by Theorem 13.

The quintic orbit

For $\ell = 5$,

$$(5, 2, 7) \longmapsto (15{,}125, 1{,}682, 16{,}807),$$

where

$$E_0 = 11 \cdot 29.$$

The next member has

$$\begin{aligned} a_2 &= 1{,}997{,}240{,}239{,}809{,}753{,}125, \\ b_2 &= 1{,}339{,}071{,}379{,}424{,}155{,}147{,}682, \\ c_2 &= 1{,}341{,}068{,}619{,}663{,}964{,}900{,}807 = 7^{25}, \end{aligned}$$

and

$$\frac{c_2}{R_2} = 29.2870\ldots$$

Here

$$E_1 = 199 \cdot 11{,}549 \cdot 892{,}254{,}749,$$

and every displayed prime is congruent to

$$\pm 1 \pmod{25}.$$

Open directions

The exact identity converts the asymptotic problem into one about repeated prime factors in a sequence of pairwise coprime cyclotomic values.

Conjecture 21: Subpower repeated-prime growth

For every admissible seed and odd prime ℓ,

$$\log W_n = o(\ell^n).$$

Equivalently, the quality of every prime-degree Chebyshev orbit tends to one.

By Proposition 14, this is an averaged sparsity statement for cyclotomic Wieferich lifts, not merely a primitive-divisor problem. Theorem 13 supplies a strong level restriction on every prime involved:

$$p \equiv \left(\frac{D_K}{p}\right) \pmod{\ell^{j+1}} \qquad (p \mid E_j).$$

A plausible first target is the weaker estimate

$$\sum_{j<n} \sum_{p^2 \mid E_j} \log p = o(\ell^n),$$

followed by control of higher valuations.

Another natural problem is to study Conjecture 21 on average over admissible seeds, where large-sieve or Chebotarev methods may be more accessible than for one fixed orbit.

The prime-degree restriction is arithmetically meaningful. It makes the new Lucas factor a single homogeneous cyclotomic value

$$\Phi_{\ell^{n+1}}(\alpha, \beta),$$

and the only systematic overlap between generations is the transfer prime ℓ, removed in the definition of E_n. Composite degrees lead to several cyclotomic layers at each step and require a different normalization.

Novelty boundary

The following distinctions are important.

The identity

$$aS_\ell(a,b)^2 + bC_\ell(a,b)^2 = (a+b)^\ell$$

is a direct form of de Moivre's formula and is not presented as a new polynomial identity.

Polynomial *abc*-transfers and sharp polynomial triples are well established; representative sources are Martin-Miao and van der Horst.

Prime-power cyclotomic factors and Lucas primitive-divisor phenomena are classical. Bilu, Hanrot, and Voutier prove the general primitive-divisor theorem for Lucas and Lehmer numbers. Alecci, Miska, Murru, and Romeo give a general definition and complete p-adic valuation theory for Lucas atoms; Ross, Shen, and Cai prove further cyclotomic congruences for Lucas sequences. The local order, valuation, and congruence statements above are specializations, not new local valuation theory.

Ratliff, Rush, and Shah construct pairwise radical-preserving families from homogeneous cyclotomic polynomials in a commutative-algebra setting. That is relevant support-separation prior art; the statement here identifies the two particular towers as coordinate multipliers in one additive orbit and derives its arithmetic dynamics.

Kym's valuation-separation theorem concerns products of Lucas terms at pairwise coprime indices. Here the atom indices are nested powers,

$$\ell^{n+1} \quad \text{and} \quad 2\ell^{n+1};$$

separation instead comes from the transfer normalization and the two coordinate branches.

The proposed contribution is the orbit-level combination:

> A complete branchwise prime genealogy, the realization as two interleaved nested-index atom towers, the radical telescope, strict monotonicity, the if-and-only-if quality criterion, the coupling of known local valuations to one global *abc*-defect term, and the exact simultaneous realization of arbitrary finite compatible prime genealogies.

A bounded search through transfer catalogues, Chebyshev arithmetic, Lucas-sequence literature, and arithmetic-dynamical primitive-divisor work did not locate these statements as an existing theorem. Specialist review is still required before making an unconditional priority claim.

AI-use statement

ChatGPT Pro was used to generate the initial proposal, including mathematical exploration, proof development, symbolic and numerical checks, literature retrieval, drafting, and the first version of the verification software. OpenAI Codex performed a subsequent proof and software audit, expanded the regression tests, and revised the proof exposition. An independent GPT-5.5 model then reviewed the core proof chain. The named author must independently verify every theorem, proof, reference, novelty statement, and computation before circulation or submission. The AI systems are not authors.

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